Appendix B. Other Abbreviations

ap	as purchased
ARS	Agricultural Research Service
DFE	Dietary Folate Equivalent
dia	diameter
DRI	Dietary Reference Intakes
fl oz	fluid ounce
FNDDS	USDA Food and Nutrient Database for Dietary Studies
g	gram
INFOODS	International Network of Food Data Systems
IU	International Unit
kcal	kilocalorie
kJ	kilojoule
lb	pound
mg	milligram
µg, mcg	microgram
ml	milliliter
NDB	Nutrient Databank
NDBS	Nutrient Databank System
NDL	Nutrient Data Laboratory
NFNAP	National Food and Nutrient Analysis Program
NLEA	Nutrition Labeling and Education Act
oz	ounce
RAE	Retinol Activity Equivalent
RE	Retinol Equivalents
RDA	Recommended Dietary Allowances, a Dietary Reference Intake
SR	USDA National Nutrient Database for Standard Reference
UL	Tolerable Upper Intake Level, a Dietary Reference Intake

Appendix C. Cooking Glossary for Meat and Poultry Items

- **Baked or Roasted:** Food cooked uncovered in an oven with no liquid added, thereby surrounding it with dry heat
- **Braised:** Food cooked on top of the range or in the oven, tightly covered in a small amount of liquid at low heat for a lengthy period of time.
- **Broiled or Grilled:** Food cooked directly under or above the heat source. Food can be broiled in an oven under a gas or electric heat source, or grilled directly over charcoal or other heat source. The term “barbecued” is sometimes used synonymously with grilled.
- **Dry heat:** A cooking technique in which heat is transferred to the food without the use of a liquid. Examples of dry heat techniques are baking, roasting, and grilling.
- **Fast Fried:** Food cooked in uncovered skillet at moderate to high heat, 3-4 minutes per side depending on thickness of the cut.
- **Fast Roasted:** Food roasted or fan-baked in an oven, such as a convection oven, for a short time at high temperature (approximately 15 minutes/500g piece of meat).
- **Fried in deep fat or oil**: Food which is cooked by immersing in hot fat or oil deep enough to completely cover the food being cooked. Average fat temperature for deep frying is 375°F/190°C, but a recipe may specify a different temperature, according to the characteristics of the food.
- **Microwaved**: Food heated or cooked using a specific type of oven that produces high frequency electromagnetic radiation as the heat source.
- **Moist heat**: A cooking technique in which heat is transferred to the food by using liquid or steam. Examples of moist heat techniques are braising, simmering, and steaming.
- **Pan-fried (fried in pan), Sautéed, or Stir fried:** Food cooked in fat which does not cover the food. Sautéing is often used to describe a method which is faster and uses less fat than pan-frying. Stir-frying is to quickly cook small pieces of food in a pan with a large surface area, using a minimum amount of fat and very high heat, while constantly stirring the food.
- **Pan-broiled:** Food cooked in uncovered skillet or frying pan over direct heat, using little or no fat. Drippings are poured off as they form.
- **Pan-browned:** Food cooked in uncovered skillet or frying pan over direct heat to obtain a brown surface on the food.
- **Poached, Simmered, or Stewed:** Food cooked in liquid at a temperature low enough that tiny bubbles just begin to break the surface (~185°F/85°C). A food being stewed involves simmering for a long period of time in a tightly covered pot.
- **Raw**: Food item in its natural state: not processed, refined, or cooked.
- **Slow Roasted:** Food roasted or fan-baked in an oven, such as a convection oven, for approximately 25-30 minutes/500g piece of meat.
- **Thawed**: Frozen food that has been exposed to a temperature higher than freezing so that it has defrosted and reached a softened state.

Appendix D. Imputing Less Than Measurements

Prepared by: Charles R. Perry, Jr. and Daniel G. Beckler, For: the Nutrient Data Laboratory, Agricultural Research Service, United States Department of Agriculture

Nutrient analysis techniques sometimes result in a less than value being reported to NDL or in the scientific literature. Such values are usually identified as either *less than the limit of quantification*, or *less than the limit of detection*. This paper presents a coherent set of rules for imputing these *less than* measurement.

NDL also considered the possibility of applying different imputation methods for nutrients that have positive dietary effects and those that have negative dietary effects. However, it is not always clear how to differentiate which nutrients are positive and those that are negative. This is complicated by the fact that certain nutrients have positive dietary effects only when they are present in very limited quantities. The working paper *Imputing Values for Trace and Not Detected Measurements* by Perry and Beckler contain separate imputation methods for positive and negative nutrients. NDL may readdress the merit of using different imputation strategies at a later date when more nutrient information is known.

There are three basic assumptions at work:

Let: *C* be the lower limit of the amount that can be present, *D* be the limit of detection (LOD), and *Q* be the limit of quantification (LOQ).

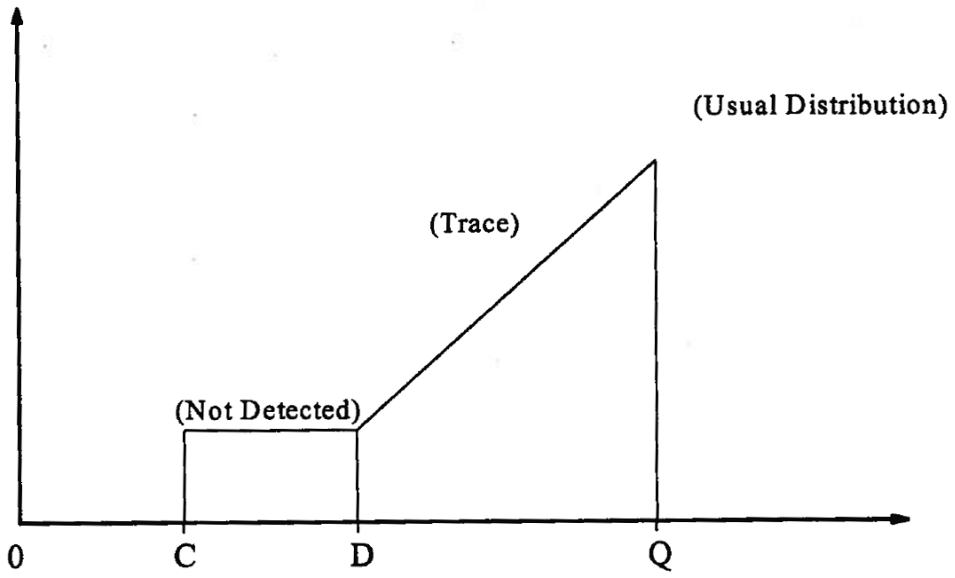


Figure 1: Assumed Distribution of Amount Present

Assumption 1: When we detect less than the limit of quantification amounts, we assume that our ability to detect the chemical is proportional to the amount present. This implies that the actual amount present T has a triangular distribution from limit of detection to the limit of quantification. Thus, we are lead to impute less than the limit of quantification amounts as:

a. $\sqrt{\frac{D^2+Q^2}{2}}$, the median of the triangular distribution over D to Q when Q and D are known;

b. $\sqrt{\frac{C^2+Q^2}{2}}$, the median of the triangular distribution over C to Q when Q and C are known and D is unknown, and

c. $\sqrt{\frac{0^2+Q^2}{2}}$, the median of the triangular distribution over 0 to Q when Q is known and C and D are unknown.

Assumption 2. When the analysis process indicates the amount of the nutrient present is less than the limit of detection, we assume that the actual amount present, T, is equally likely to be located at any point in the range of values where the actual values are known to lie. This implies that the actual amount present, T, has a uniform distribution from the lower limit of the amount present to the limit of detection. Thus, we are lead to impute less than the limit of detection amounts as:

a. $\frac{C+D}{2}$, the median of the uniform distribution over C to D when C and D are known;

b. $\frac{0+D}{2}$ the median of the uniform distribution over O to D when C is unknown and D is known;

c. $\frac{C+C}{2}$ = C when C is known and D is unknown;

d. $\frac{0+0}{2}$ = 0 when C and D are both unknown.

Assumption 3: When the limit of quantification is not known for a set of measurements, we first impute it from the current data measurements (or from a similar set of measurements) by assuming its actual value is uniformly distributed between its maximum likelihood estimate from the current measurements, min(current quantifiable values), and the next lower value that is known. That is:

a. $Q = \frac{1}{2}(\min(\text{current measurements}) + D)$, when D is known;

b. $Q = \frac{1}{2}(\min(\text{current measurements}) + C)$, when D is unknown and C is known;

c. $Q = \frac{1}{2}\min(\text{current measurements})$, when both C and D are unknown.

This approach to imputing missing Q is likely to be somewhat conservative and result in a lower values for Q than the actual limit of quantification.

We can summarize the decision rules for imputation that result from the above assumptions as follows:

Case 1: C, D and Q are known	**Impute Trace As:**	**Impute Not Detected As:**
When *More* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{C + D}{2}$

Case 2: D & Q known, C unknown	**Impute Trace As:**	**Impute Not Detected As:**
When *More* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{0 + D}{2}$

Case 3: C & Q known, D unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{C^2 + Q^2}{2}}$	C

Case 4: Q known, C & D unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{0^2 + Q^2}{2}}$	0

Case 5: C & D known, Q unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{C + D}{2}$

Case 6: D known, Q &C unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{0 + D}{2}$

Case 7: C known, D & Q unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{C^2 + Q^2}{2}}$	C

Case 8: D, Q & C unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{0^2 + Q^2}{2}}$	0

References for Appendix D

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Appendix E. Imputing Values for Trace and Not Detected Measurements.

Prepared by: Charles R. Perry, Jr. and Daniel G. Beckler for Nutrient Data Laboratory, Agricultural Research Service, United States Department Of Agriculture

Nutrient analysis techniques sometimes result in a trace or undetectable amounts being reported to NDL or in the scientific literature. This paper presents a coherent set of rules for imputing when the measurement is either a trace or not detected.

There are three basic assumptions at work:

Let: *C,* be the lower limit of the amount that can be present; D*,* be the limit of detection and *Q* be the limit of quantification.

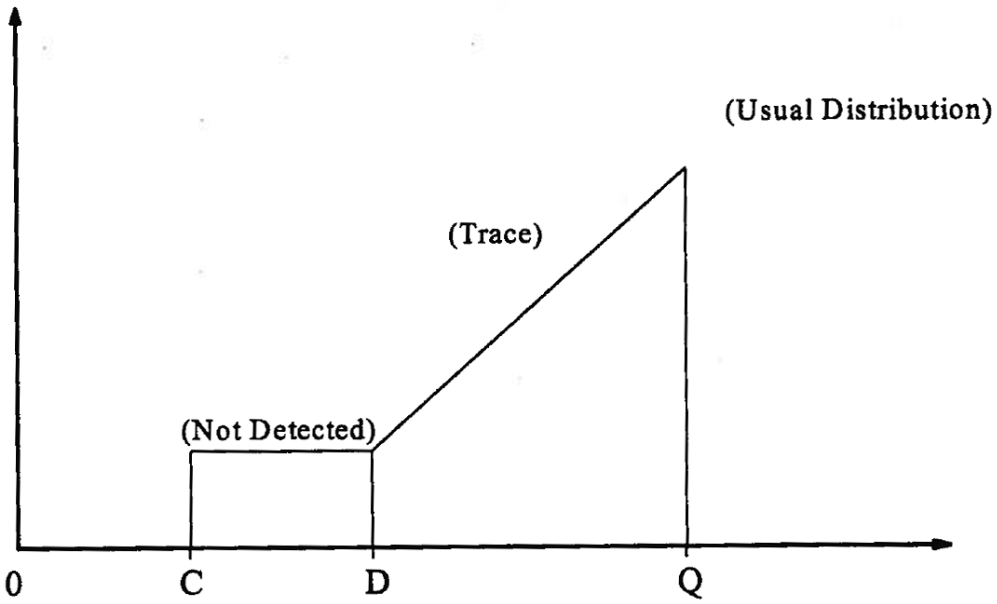


Figure 1: Assumed Distribution of Amount Present

Assumption 1: When we detect a trace, we assume that our ability to detect the chemical is proportional to the amount present. This implies that the actual amount present T has a triangular distribution from limit of detection to the limit of quantification. Also, since the dietary effect of a nutrient can be either positive or negative, we are lead to act conservatively and impute trace measurements conditionally on the nutrient's dietary effect.

When the dietary effect of more of the nutrient is positive, trace measurements should be imputed as:

a. $\sqrt{\frac{D^2 + Q^2}{2}}$, the median of the triangular distribution over D to Q when Q and D are known;

b. $\sqrt{\frac{C^2 + Q^2}{2}}$, the median of the triangular distribution over C to Q when Q and C are known and D is unknown, and

c. $\sqrt{\frac{0^2 + Q^2}{2}}$, the median of the triangular distribution over 0 to Q when Q is known and C and D are unknown.

When the dietary effect of less of the nutrient is positive, trace measurements should be imputed as:

a. $\sqrt{\frac{D^2 + Q^2}{2}}$, when Q and D are known;

b. Q, when Q and C are known and D is unknown, and

c. Q, when Q is known and C and D are unknown.

Assumption 2. When we do not detect the chemical, we assume that the actual amount present, T, is equally likely to be located at any point in the range of values where the actual values are known to lie. This implies that the actual amount present, T, has a uniform distribution from the lower limit of the amount present to the limit of detection.

When the dietary effect of more of the nutrient is positive, trace measurements should be imputed as:

a. $\frac{C + D}{2}$, the median of the uniform distribution over C to D when C and D are known;

b. $\frac{0+D}{2}$ the median of the uniform distribution over O to D when C is unknown and D is known;

c. $\frac{C+C}{2}$ = C when C is known and D is unknown;

d. $\frac{0+0}{2}$ = 0 when C and D are both unknown.

When the dietary effect of less of the nutrient is positive, trace measurements should be imputed as:

a. $\frac{C+D}{2}$, when C and D are known;

b. D, when C is unknown and D is known;

c. $\frac{C+Q}{2}$, when C is known and D is unknown;

d. Q, when C and D are both unknown.

Assumption 3: When the limit of quantitation is not known for a set of measurements, we first impute it from the current data measurements (or from a similar set of measurements) by assuming its actual value is uniformly distributed between its maximum likelihood estimate from the current measurements, min(current quantitation values), and the next lower value that is known.

When the dietary effect of more of the nutrient is positive, trace measurements should be imputed as:

a. $Q = \frac{1}{2}(\min(\text{current measurements}) + D)$, when D is known;

b. $Q = \frac{1}{2}(\min(\text{current measurements}) + C)$, when D is unknown and C is known;

c. $Q = \frac{1}{2}\min(\text{current measurements})$, when both C and D are unknown.

When the dietary effect of less of the nutrient is positive, trace measurements should be imputed as: $Q = \min(\text{current measurements})$, regardless of knowledge C and D.